

Appendix 1: Weierstrass

Let $\pi : E \rightarrow S$ be an elliptic curve over $\mathbb{Q}$, with identity e . Define the *Hodge line bundle* $\lambda := \omega_{S/(S/E)}[-1]$,¹ and put $V_n := \pi_* \mathcal{O}_E(ne)$ for $n \geq 1$.

Lemma 1. *There are a natural fibre sequence and a canonical decomposition*

$$\mathcal{O}_S \xrightarrow{u} V_2 \xrightarrow{p} \lambda^{-2}, \quad V_3 \simeq V_2 \oplus \lambda^{-3}.$$

The second summand is the anti-invariant summand for inversion.

Proof. We have $V_1 \simeq \mathcal{O}_S$ and $e^* \mathcal{O}_E(e) \simeq \lambda^{-1}$. Pushing forward the divisor sequences gives $V_{n-1} \rightarrow V_n \rightarrow \lambda^{-n}$ for $n \geq 2$. Write $V_n^\pm$ for the summands under inversion, which acts on λ^{-n} by $(-1)^n$. Thus $V_2^- = 0$, $V_3^+ \simeq V_2$, and $V_3^- \simeq \lambda^{-3}$. $\square$

Let $y : \lambda^{-3} \rightarrow V_3$ be the inclusion supplied by the lemma. Let $f : \mathcal{X} \rightarrow S$ be the sheaf of splittings of p , with universal map

$$x : f^* \lambda^{-2} \longrightarrow f^* V_2, \quad (f^* p)x \simeq \text{id}.$$

The sheaf $\mathcal{X}$ is a torsor for $\mathbf{V}_S(\lambda^2)$. We use the natural products $V_m \otimes V_n \rightarrow V_{m+n}$.

Put $\mathcal{W} = \mathcal{O}_S \oplus \lambda^2 \oplus \lambda^3$ and $\varpi : \mathbb{W} = \mathbb{P}_S(\mathcal{W}) \rightarrow S$. Let $\mathcal{T} = \mathcal{O}_{\mathbb{W}}(-1)$ and let $\sigma_i : \mathcal{T} \rightarrow \varpi^* \lambda^i$, for $i = 0, 2, 3$, be the three components of $\mathcal{T} \rightarrow \varpi^* \mathcal{W}$.

Proposition 2 (Universal Weierstrass embedding). *Over $\mathcal{X}$, the maps $(1, x, y)$ identify $V_3 \simeq \mathcal{O} \oplus \lambda^{-2} \oplus \lambda^{-3}$. There are canonical sections $a_i(x) : \mathcal{O}_{\mathcal{X}} \rightarrow \lambda^i$, for $i = 2, 4, 6$, such that evaluation identifies $E_{\mathcal{X}}$ with the cubic in $\mathbb{W}_{\mathcal{X}}$*

$$\sigma_3^{\otimes 2} \sigma_0 = \sigma_2^{\otimes 3} + a_2(x) \sigma_2^{\otimes 2} \sigma_0 + a_4(x) \sigma_2 \sigma_0^{\otimes 2} + a_6(x) \sigma_0^{\otimes 3}. \quad (1)$$

Proof. The maps $1, x, x^2, x^3$ give a filtered equivalence

$$(1, x, x^2, x^3) : \mathcal{O} \oplus \lambda^{-2} \oplus \lambda^{-4} \oplus \lambda^{-6} \xrightarrow{\sim} V_6^+.$$

The last component of y^2 is canonically $\text{id}_{\lambda^{-6}}$; the remaining components define a_6, a_4, a_2 , with $y^2 \simeq x^3 + a_2 x^2 + a_4 x + a_6$.

A direct check shows that evaluation is an isomorphism onto this cubic. $\square$

Proposition 3 (Canonical normalization). *The coefficient map $a_2 : \mathcal{X} \rightarrow \mathbf{V}_S(\lambda^2)$ is an equivalence. Its fibre over zero gives a canonical splitting x_0 of p and hence*

$$V_3 \simeq \mathcal{O}_S \oplus \lambda^{-2} \oplus \lambda^{-3}.$$

Put $a_i := a_i(x_0)$ for $i = 4, 6$. There is a natural embedding $E \hookrightarrow \mathbb{W}$, with equation

$$\sigma_3^{\otimes 2} \sigma_0 = \sigma_2^{\otimes 3} + \varpi^* a_4 \sigma_2 \sigma_0^{\otimes 2} + \varpi^* a_6 \sigma_0^{\otimes 3}. \quad (2)$$

The identity section is induced by $\lambda^3 \subset \mathcal{W}$. Every construction commutes with base change.

Proof. Clear from above. $\square$

¹Jacob Lurie, *Elliptic Cohomology I: Spectral Abelian Varieties*, preprint, February 3, 2018, §5.2.